

Bubble Sort Rounds I

Time limit: 1.00 s

Memory limit: 512 MB

Bubble sort is a sorting algorithm that consists of a number of rounds. On each round the algorithm scans the array from left to right and swaps any adjacent elements that are in the wrong order.

Given an array of n integers, calculate the number of bubble sort rounds needed to sort the array.

Input

The first line has an integer n : the array size.

The next line has n integers $x_1, x_2, \dots, x_n$: the array contents.

Output

Print one integer: the number of rounds.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input	Output
5 3 2 4 1 4	3

Explanation: Bubble sort needs three rounds to sort this array. The array contents after each round are $[2, 3, 1, 4, 4]$, $[2, 1, 3, 4, 4]$, and $[1, 2, 3, 4, 4]$.